13. Conclusion

This project applied a comprehensive data analytics approach to investigate Airbnb market performance and identify factors influencing pricing and investment opportunities.

Using exploratory data analysis, correlation analysis, machine learning modelling, SHAP interpretation, and interactive Power BI dashboards, the project successfully transformed large-scale Airbnb listing and review data into meaningful business insights.

The analysis identified that **location is the most important factor influencing Airbnb performance**. Geographic characteristics, particularly neighbourhood-level differences, strongly affect pricing potential and customer demand. This highlights the importance of selecting the right market location before making investment decisions.

The machine learning analysis further confirmed that **property characteristics, especially bedrooms and accommodation capacity, are key drivers of listing prices**. Larger properties generally achieve higher prices due to their ability to attract families, groups, and higher-value customers. Among the tested models, Random Forest achieved the best predictive performance, demonstrating the effectiveness of machine learning in estimating Airbnb pricing outcomes.

Demand analysis revealed that Airbnb activity is concentrated in specific cities and neighbourhoods, with seasonal patterns influencing customer behaviour. This demonstrates the importance of considering both geographic attractiveness and demand stability when evaluating investment opportunities.

For **investors**, the findings suggest that capital should be allocated towards high-demand neighbourhoods with strong revenue potential, suitable property characteristics, and evidence-based price predictions.

For **existing Airbnb hosts**, recommendations focus on improving revenue through dynamic pricing strategies, enhancing guest experience, maintaining strong reviews, and continuously monitoring market trends.

Overall, this project demonstrates how data analytics can support smarter Airbnb decision-making by combining descriptive analytics, predictive modelling, and business intelligence. The developed framework provides stakeholders with a data-driven approach to identify profitable markets, understand pricing factors, optimise listing performance, and improve investment decisions.

Future improvements involving additional data sources, advanced modelling techniques, and automated recommendation systems could further enhance the accuracy and practical value of this solution.

In conclusion, successful Airbnb performance is achieved through the combination of strategic location selection, appropriate property investment, predictive analytics, and continuous data-driven optimisation. This approach enables investors and hosts to make informed decisions, reduce uncertainty, and maximise long-term revenue opportunities in the competitive short-term rental market.